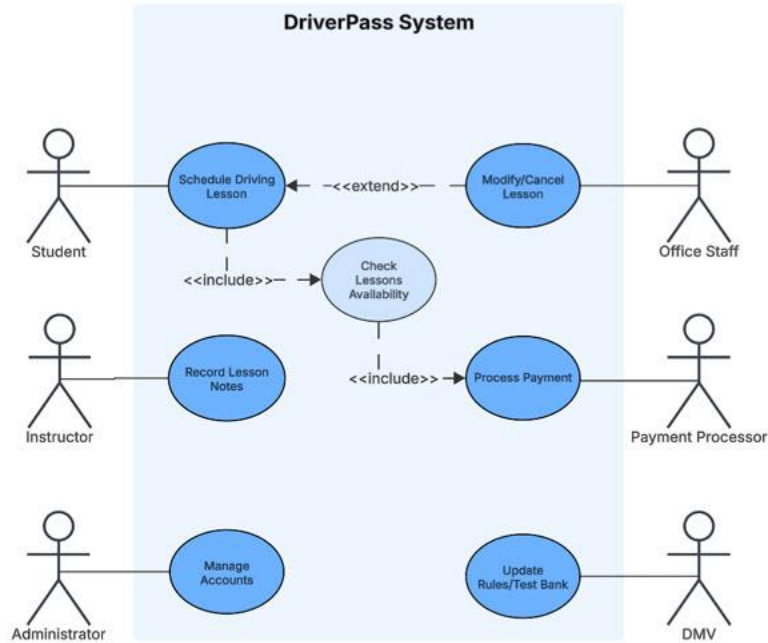


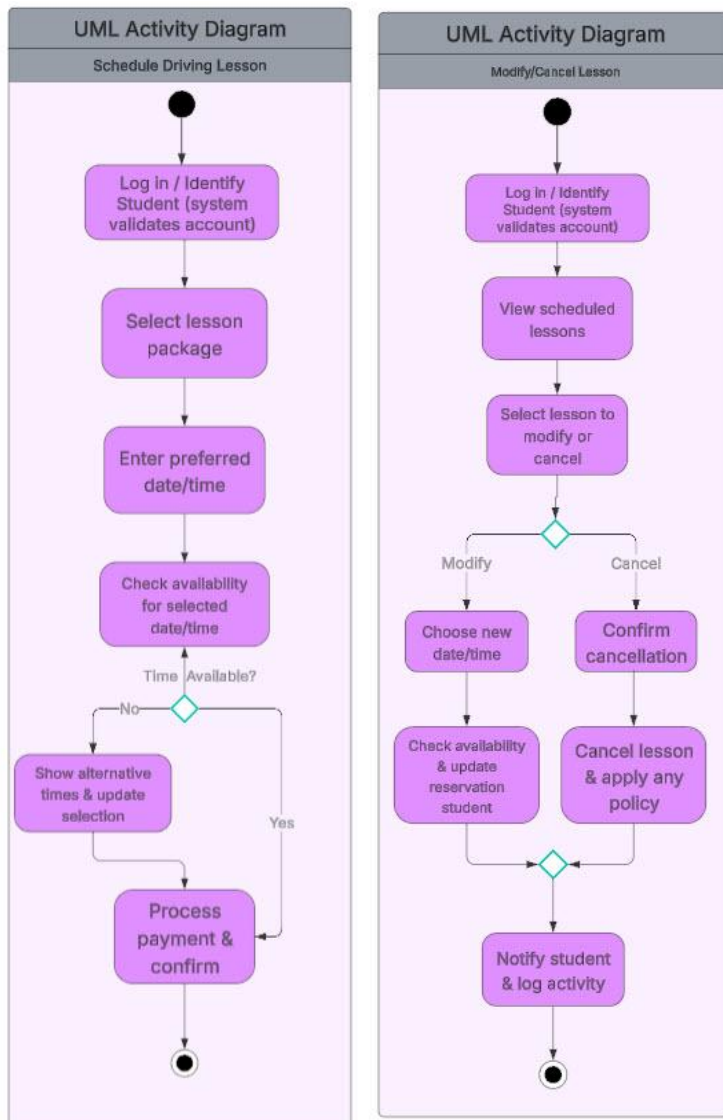
CS 255 System Design Document

UML Diagrams

UML Use Case Diagram

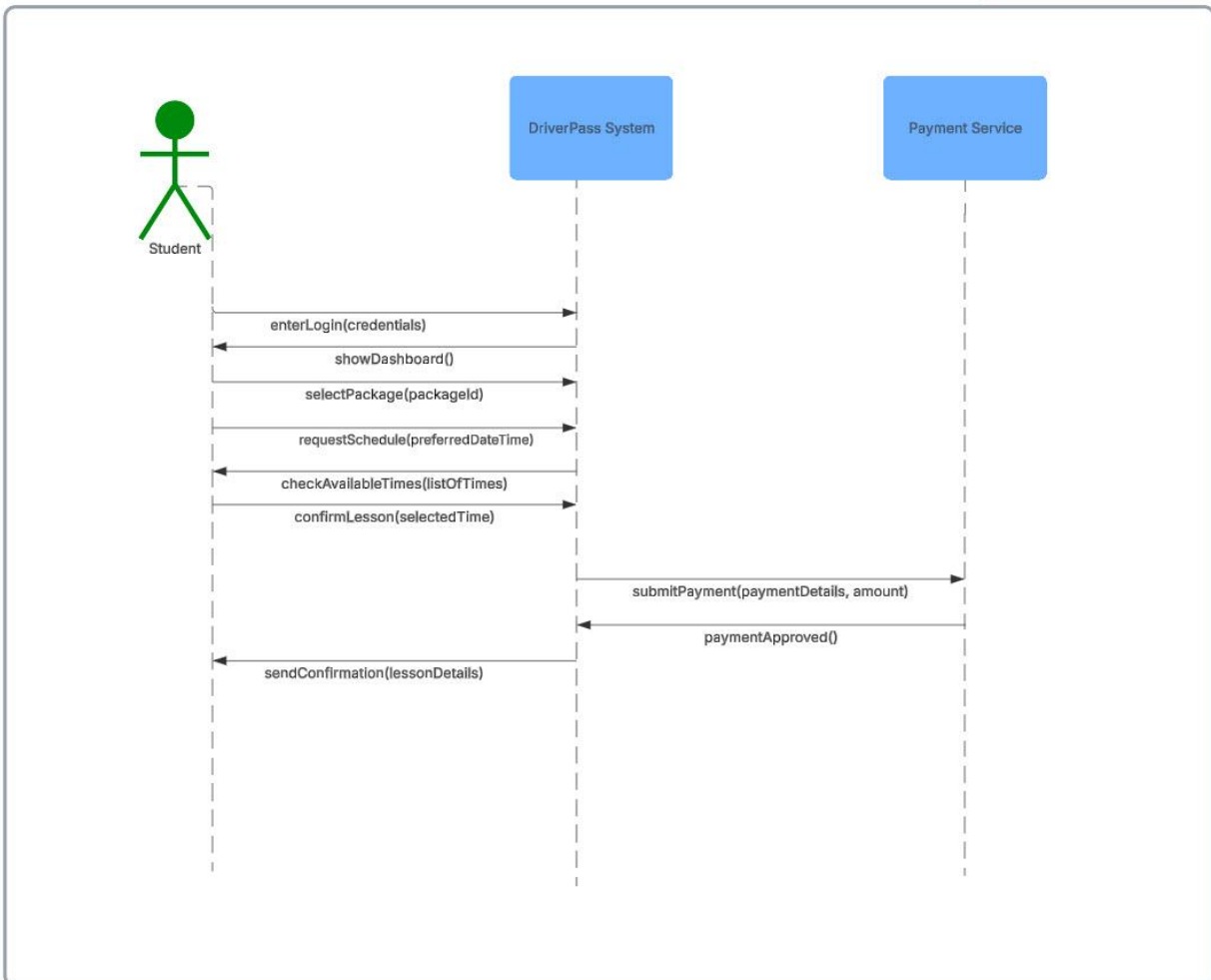


UML Activity Diagrams

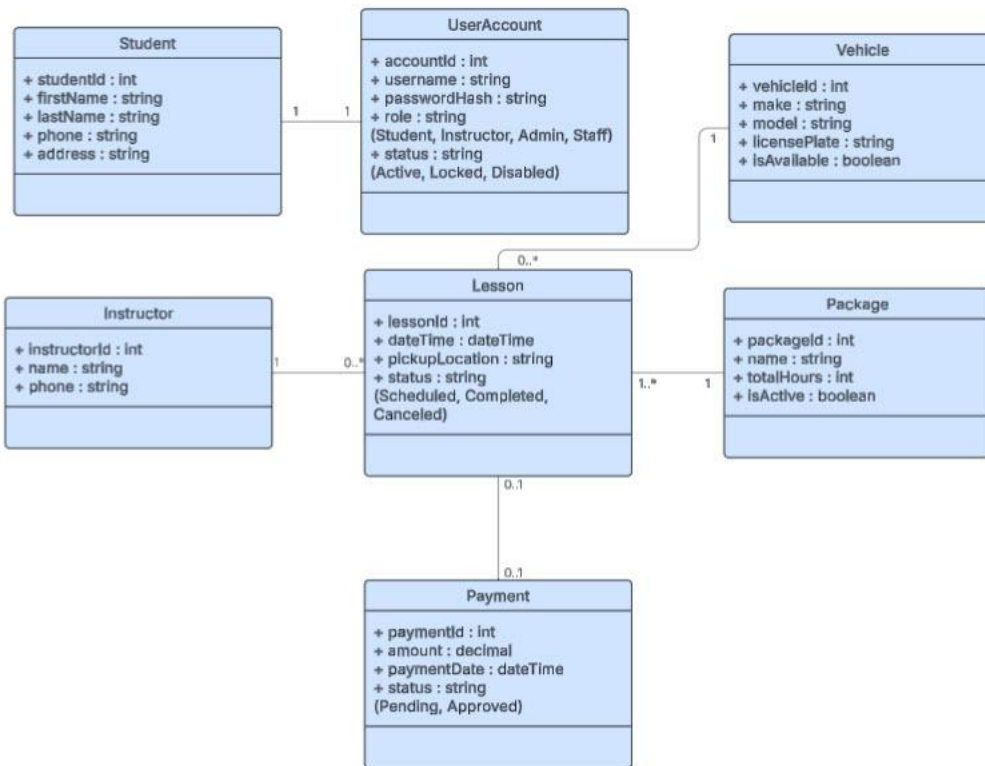


UML Sequence Diagram

Schedule Driving Lesson Sequence diagram



UML Class Diagram



Technical Requirements

The functional and nonfunctional requirements from my Project One document guide the technical needs of the DriverPass system. Because the system must let students create accounts, log in, schedule lessons, modify or cancel existing lessons, take online courses, and make payments, the solution requires a reliable web server that can host the application and support secure Hypertext Transfer Protocol Secure (HTTPS) communication. A separate database server is needed to store user accounts, lessons, instructors, vehicles, packages, and payment records, since these were key data elements identified in the functional requirements. To support these features, the system will use an operating system such as Microsoft Windows Server or Linux, along with a modern server-side framework such as Java Spring Boot or Microsoft .NET to handle the workflows shown in the activity and sequence diagrams. A relational Database Management System like My Structured Query Language (MySQL) or PostgreSQL is required to manage all stored information. Because one of the functional requirements is to process payments securely, the system must also integrate with a third-party payment provider through a secure Application Programming Interface connection.

Nonfunctional requirements such as system security, reliability, and performance also affect the technical setup. The system must use Secure Sockets Layer or Transport Layer Security certificates to encrypt sensitive information, password hashing for account protection, and role-based access control to separate student, instructor, staff, and administrator permissions. Regular backups and a disaster recovery plan are required to ensure availability. To build and maintain the system, the development team will need tools such as Lucidchart for Unified Modeling Language diagrams and GitHub or GitLab for version control. The overall system can be hosted either in the cloud, using Amazon Web Services, Microsoft Azure, or Google Cloud Platform, or on an on-premise server as long as it supports scalability.

as the number of students and instructors grows. These technical requirements ensure the system can meet all the functional and nonfunctional expectations outlined in Project One.